

A Demonstration of Morphing Identity: Exploring Self-Other Identity Continuum through Interpersonal Facial Morphing

Kye Shimizu

Sony Computer Science Laboratories, Inc.
Tokyo, Japan
kyeshimizu@gmail.com

Jun Nishida

Sony Computer Science Laboratories, Inc.
University of Maryland, College Park
Japan/United States
jun.nishida@acm.org

Santa Naruse

Sony Computer Science Laboratories, Inc.
Tokyo, Japan
santanaruse@gmail.com

Shunichi Kasahara

Sony Computer Science Laboratories, Inc.
Okinawa Institute of Science and Technology
Graduate University
Japan
kasahara@csl.sony.co.jp

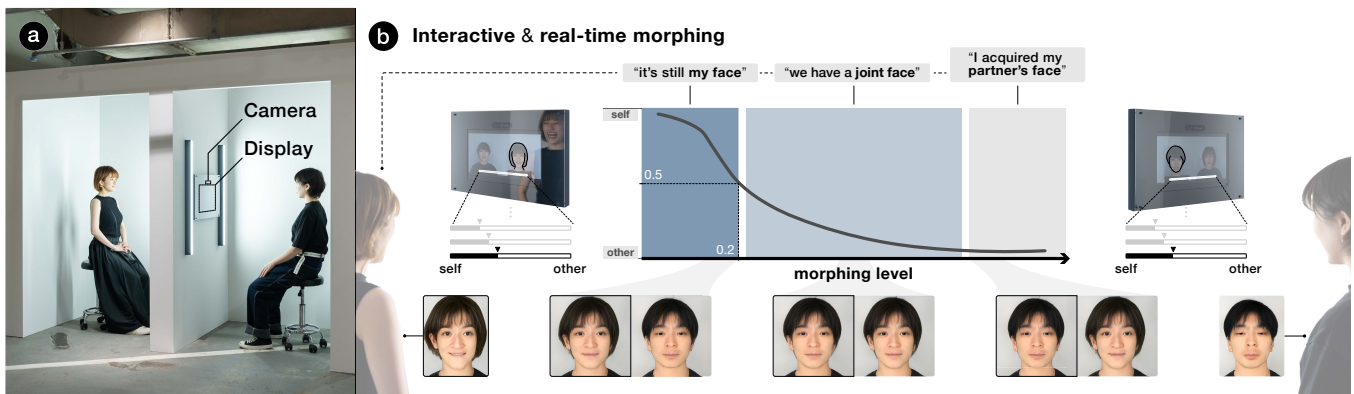


Figure 1: (a) We developed Morphing Identity, an interactive system that blends facial images of two users in real-time. Users' facial images are gradually blended from images of self to that of partner, while both users can see and interact with each other. (b) We installed this system in several exhibitions to conduct a qualitative study to explore interpersonal experiences and quantitative analysis to provide further insights into the boundary between self and others.

ABSTRACT

We explored continuous changes in self-other identity by designing an interpersonal facial morphing experience where the facial images of two users are blended and then swapped over time. To explore this with diverse social relationships, we conducted qualitative and quantitative investigations through public exhibitions. We found that there is a window of self-identification as well as a variety of interpersonal experiences in the facial morphing process.

From these insights, we synthesized a Self-Other Continuum represented by a sense of agency and facial identity. This continuum has implications in terms of the social and subjective aspects of interpersonal communication, which enables further scenario

design and could complement findings from research on interactive devices for remote communication.

ACM Reference Format:

Kye Shimizu, Santa Naruse, Jun Nishida, and Shunichi Kasahara. 2023. A Demonstration of Morphing Identity: Exploring Self-Other Identity Continuum through Interpersonal Facial Morphing. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Emerging Technologies (SIGGRAPH '23 Emerging Technologies)*, August 06–10, 2023, Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3588037.3595394>

1 INTRODUCTION

The psychology community has been exploring facial identity boundaries by transferring the sense of ownership to a different person's face by means of a visual and tactile stimulation technique [Tajadura-Jiménez et al. 2012] in the static situation. To explore this in a more interactive situation, we built a system that morphs two users' facial identities together in real-time while allowing them to communicate through verbal and non-verbal factors (i.e., facial expressions, body actions), which play a critical role in

creating a sense of motor agency. Our system can decompose and re-compose the relationship between one's own real-time movements (facial signals) and facial identity using GAN image generation [Karras et al. 2019] and projection methods [Abdal et al. 2019], as well as real-time facial reenactment [Siarohin et al. 2019] and alignment using frames from the morphing sequence. This allows us to create a multifaceted relationship in interpersonal communication where the user is able to become the partner's face and maneuver it, while at the same time, the partner is able to move the user's face, providing further interaction design implications for face-changing applications [Shimizu et al. 2023].

2 IMPLEMENTATION

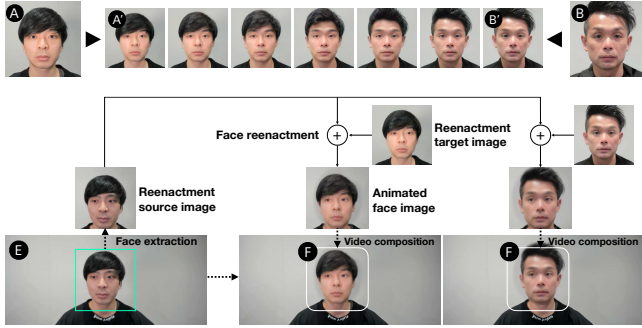


Figure 2: Overview of the morphing process

First, the participant's face was captured, the system cropped the facial areas from the camera image and utilized the server computer to search for a latent vector corresponding to the facial image by projecting the image into the latent space of the GAN. We employed a pre-trained model based on StyleGAN2 [Karras et al. 2019].

Face morphing can then be performed by mixing the latent vectors of two face images with $W'_f = W_f + r(W_t - W_f)$. Here, W_f and W_t represent the latent vector of the original facial image (from) and the destination facial image (to).

After 100 morphed images were generated, the display showed each participant's real-time moving faces (i.e. 100 percent themselves). The real-time face morphing started to proceed with a morphing speed of (2%/1 sec). During the face morphing, the display showed both faces of the participants in side by side. To allow interaction between users while the facial morphing proceeds, face reenactment technology [Siarohin et al. 2019] was integrated to enable the transfer of a facial expression and head pose of source input to be transferred to a target face while preserving the target facial identity.

3 FINDINGS AND IMPLICATIONS

From our findings [Shimizu et al. 2023], we found that participants recognized the morphed faces as their own until the 20% of interpersonal morphing level, suggesting a wider range of self-identification. We then synthesized a continuum that represents the experiences with interpersonal face morphing, as depicted in Figure 3. It consists of three concepts of facial identity which are grouped based on how users recognize the facial identity and the feeling of agency over

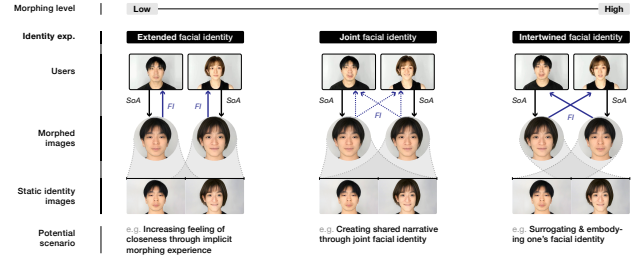


Figure 3: Self-Other Continuum: consisting of three concepts of facial identity (SoA: sense of agency, FI: facial identity)

the facial images: 1) **extended facial identity**, where self-identity is implicitly extended to that of a partner, 2) **joint facial identity**, where their facial identities are mixed, but each user still feels an essence of each identity, and 3) **intertwined facial identity**, where their facial identities are swapped.

4 VISITOR EXPERIENCE

Through engaging in real-time interactive face morphing while conversing and viewing realistic facial expressions, visitors can explore their own boundaries between themselves and others at the venue. They can also experience joy and embarrassment through controlling and/or being controlled by their partners. By sharing a common facial identity with their participating partner, similarities, and differences become apparent, creating a shared context for conversations that can lead to topics such as family connections and characteristic traits, as seen in the qualitative study.

5 CONCLUSION

Through the design of an interpersonal facial morphing experience, we investigated continuous changes in self-other identification and various interpersonal experiences. We then synthesized a Self-Other Continuum, symbolized by a sense of agency and facial identity, which would enhance remote communication setups.

ACKNOWLEDGMENTS

This work was supported by the JST Moonshot R&D Program (Grant Number JPMJMS2013).

REFERENCES

- Rameen Abdal, Yipeng Qin, and Peter Wonka. 2019. Image2StyleGAN: How to Embed Images Into the StyleGAN Latent Space?. In *2019 IEEE/CVF International Conference on Computer Vision (ICCV)*. 4431–4440.
- Tero Karras, Samuli Laine, Miika Aittala, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. 2019. Analyzing and Improving the Image Quality of StyleGAN.
- Kye Shimizu, Santa Naruse, Jun Nishida, and Shunichi Kasahara. 2023. Morphing Identity: Exploring Self-Other Identity Continuum through Interpersonal Facial Morphing Experience. In *Proceedings of the 2023 CHI Conference on Human Factors in computing systems (CHI '23)* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA.
- Aliaksandr Siarohin, Stéphane Lathuilière, Sergey Tulyakov, Elisa Ricci, and Nicu Sebe. 2019. First Order Motion Model for Image Animation. In *Conference on Neural Information Processing Systems (NeurIPS)*.
- Ana Tajadura-Jiménez, Stephanie Grehl, and Manos Tsakiris. 2012. The other in me: interpersonal multisensory stimulation changes the mental representation of the self. *PLoS One* 7, 7 (July 2012), e40682.